Principles of Statistics

Part II

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Principles of Statistics

0. INTRODUCTION

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and $X : \Omega \to \mathbb{R}$ a random variable with distribution function

$$F(t) = \mathbb{P}\big(\{\omega : X(\omega) \leq t\}\big) = \begin{cases} \displaystyle\int_{-\infty}^{t} f(x)\,dx, & \text{if } X \text{ is continuous with pdf } f, \\ \displaystyle\sum_{x \leq t} f(x), & \text{if } X \text{ is discrete with pmf } f. \end{cases}$$

We denote $P = \mathbb{P} \circ X^{-1}$, the law or "distribution" of X.
We observe $X_1, \ldots, X_n$, which are iid copies of X, where n is the sample size.

Def 1. A statistical model is a collection $\{f_\theta : \theta \in \Theta\}$ of pdfs or pmfs, or $\{P_\theta : \theta \in \Theta\}$ of laws, indexed by a parameter space Θ.

Ex.

- $N(\theta, 1)$, $\Theta = \mathbb{R}$.
- $N(\mu, \sigma^2)$, $\theta = (\mu, \sigma^2)$, $\Theta = \mathbb{R} \times (0, \infty)$.
- $\text{Poi}(\theta)$, $\Theta = [0, \infty)$.

Def 2. We say the model is correctly specified when $\exists\, \theta_0 \in \Theta$ such that $P = P_{\theta_0}$.
We will refer to the "true parameter" as θ_0.

Goals of Statistics

1. Estimation: We define an estimator
$$\widehat{\theta}_n = \widehat{\theta}_n(X_1, \ldots, X_n),$$
which we hope is close to θ_0 when $X_i \overset{\text{iid}}{\sim} P_{\theta_0}$, for all $\theta_0 \in \Theta$, when n is large.
2. Testing: We have a null hypothesis
$$H_0 : \theta \in \Theta_0 \subseteq \Theta,$$
and an alternative hypothesis
$$H_1 : \theta \in \Theta_1 \subseteq \Theta.$$
We define a test,
$$\psi_n := \psi_n(X_1, \ldots, X_n),$$
such that $\psi_n = 0$ if H_0 is true, and $\psi_n = 1$ if H_1 is true, w.h.p.

3. Uncertainty Quantification

- Define a confidence set $C_n(X_1, \ldots, X_n; \alpha)$, such that

$$P_\theta(\theta \in C_n) \overset{(\geq)}{=} 1 - \alpha \qquad \forall\, \theta \in \Theta,$$

  where α is the significance level.
- Define a Bayesian credible set, a region of high posterior probability.

I. LIKELIHOOD PRINCIPLE

Def 3. The likelihood function

$$L_n(\theta) = \prod_{i=1}^{n} f_\theta(X_i).$$

Note: This is a random function, as it depends implicitly on $X_1, \ldots, X_n$.

The log-likelihood function

$$\ell_n(\theta) = \sum_{i=1}^{n} \log f_\theta(X_i),$$

we write

$$\overline{\ell}_n(\theta) = \frac{1}{n}\ell_n(\theta)$$

for a normalised log-likelihood.

Def 4. A maximum likelihood estimator $\widehat{\theta}_{\text{MLE}}$ is any estimator with

$$L_n\big(\widehat{\theta}_{\text{MLE}}\big) = \max_{\theta \in \Theta} L_n(\theta).$$

Ex.

$$X_i \overset{\text{iid}}{\sim} \text{Poi}(\theta).$$

$$L_n(\theta) = \prod_{i=1}^{n} e^{-\theta} \frac{\theta^{X_i}}{X_i!}.$$

$$\ell_n(\theta) = -n\theta + (\log \theta) \sum X_i + \text{const.}$$

$$\ell_n'(\theta) = -n + \frac{\sum X_i}{\theta} \overset{\text{FOC}}{=} 0$$

$$\implies \quad \text{suggests } \widehat{\theta}_{\text{MLE}} = \frac{\sum_{i=1}^{n} X_i}{n} := \overline{X}_n \qquad \text{“sample mean”}.$$

SOC:

$$\ell_n''(\theta) = -\frac{1}{\theta^2} \sum X_i < 0 \qquad \text{if } \sum X_i > 0.$$

If $\sum X_i = 0$,

$$\ell_n(\theta) = -n\theta \quad \implies \quad \widehat{\theta}_{\text{MLE}} = 0.$$

In every case

$$\widehat{\theta}_{\text{MLE}} = \overline{X}_n.$$

Ex.

$$X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2), \qquad \theta = (\mu, \sigma^2).$$

$$\ell_n(\theta) = -\frac{n}{2}\log\pi - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2.$$

FOC:

$$\frac{\partial \ell_n}{\partial \mu} = -\frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu) = 0,$$

$$\frac{\partial \ell_n}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(X_i - \mu)^2 = 0.$$

We can check that the Hessian of ℓ_n is negative definite $\forall\, \theta \in \mathbb{R} \times (0, \infty)$.
Hence,

$$\widehat{\theta}_{\text{MLE}} = \begin{pmatrix} \overline{X}_n \\ \underbrace{\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2}_{\text{“Sample variance”}} \end{pmatrix}.$$

Non-examinable

The likelihood principle poses that if two observations

$$X = (X_1, \ldots, X_n) \qquad \text{and} \qquad X^* = (X_1^*, \ldots, X_n^*)$$

give rise to likelihoods L_n, L_n^* respectively, such that

$$L_n(\theta) = L_n^*(\theta) \cdot M \qquad \forall\, \theta \in \Theta,$$

and some constant M, then they should produce the same “inference” on the true parameter.

- The MLE, Bayes estimators, credible sets satisfy the likelihood principle.
- Confidence sets, tests generally do not.

Birnbaum (1962)

$$\text{Conditionality} + \text{Sufficiency} \quad \Longrightarrow \quad \text{Likelihood principle.}$$

end of lecture 1

II. Information Geometry

The score function is

$$S_n(\theta) = \frac{\partial}{\partial \theta}\ell_n(\theta), \qquad \Theta \subseteq \mathbb{R}^d.$$

In many cases,

$$S_n(\widehat{\theta}_{\text{MLE}}) = 0,$$

where, by convention,

$$S_n(\theta') = \frac{\partial}{\partial \theta}\ell_n(\theta)\bigg|_{\theta=\theta'}.$$

Let X be a r.v. taking values in $\mathcal{X}$, and $g : \mathcal{X} \to \mathbb{R}$. Take a model $\{f(x,\theta) : \theta \in \Theta\}$. We will write

$$\mathbb{E}_\theta g(X) = \int g(x) f(x,\theta)\, dx, \qquad X \sim f(\cdot,\theta).$$

Obs 1.

Suppose the model $\{f(\cdot,\theta) : \theta \in \Theta\}$ is such that

$$\mathbb{E}_{\theta_0} \big|\log f(X,\theta)\big| < \infty \qquad \forall \theta \in \Theta.$$

Define

$$\ell(\theta) = \mathbb{E}_{\theta_0} \log f(X,\theta), \qquad X \sim P_{\theta_0}.$$

The normalised log-likelihood

$$\bar{\ell}_n(\theta) = \frac{1}{n} \sum_{i=1}^{n} \log f(X_i,\theta)$$

can be thought of as a sample version of $\ell(\theta)$.

Suppose the model is correctly specified, such that X has pdf $f(\cdot,\theta_0)$ for some $\theta_0 \in \Theta$.

Fact. Then $\ell(\theta)$ is maximised at θ_0. This justifies using the MLE to approximate θ_0 when n is large.

Indeed,

$$\begin{aligned}
\ell(\theta) - \ell(\theta_0) &= \mathbb{E}_{\theta_0} \left(\log \frac{f(X,\theta)}{f(X,\theta_0)} \right) \\
&\le \log \mathbb{E}_{\theta_0} \left(\frac{f(X,\theta)}{f(X,\theta_0)} \right) \\
&= \log \int \frac{f(x,\theta)}{f(x,\theta_0)} f(x,\theta_0)\, dx \\
&= \log 1 = 0.
\end{aligned}$$

By Jensen's inequality, as log is concave, with equality if

$$\frac{f(X,\theta)}{f(X,\theta_0)}$$

is a.s. constant.

$$\therefore \quad \ell(\theta) \le \ell(\theta_0).$$

If

$$f(\cdot,\theta) = f(\cdot,\theta') \quad \Longrightarrow \quad \theta = \theta',$$

we say the parameter is identifiable.

In that case, if $\theta \neq \theta_0$, Jensen's inequality is strict, so θ_0 is actually the unique maximizer of ℓ.

Rmk. The difference

$$\ell(\theta_0) - \ell(\theta) = \int \log \frac{f(x,\theta_0)}{f(x,\theta)} f(x,\theta_0)\, dx$$

is called the Kullback–Leibler divergence or relative entropy between P_{θ_0} and P_θ.

A notion of distance which is not symmetric.

Obs 2.

$$0 = \frac{\partial}{\partial\theta} \underbrace{\int f(x,\theta)\,dx}_{=1\ \forall\theta} = \int \frac{\partial}{\partial\theta} f(x,\theta)\,dx$$
$$= \int \frac{\partial \log f(x,\theta)}{\partial\theta} f(x,\theta)\,dx$$
$$= \mathbb{E}_\theta\left[\frac{\partial \log f(X,\theta)}{\partial\theta}\right].$$

[There are some conditions for this. (Leibniz rule). See P & M.]
"$S_n(\theta)$ is centred when $X \sim P_\theta$."

Obs 3.

Def 5. Let $\Theta \subseteq \mathbb{R}^d$. For any $\theta \in \text{int}(\Theta)$, the Fisher Information matrix is

$$I(\theta) = \mathbb{E}_\theta\left[\frac{\partial}{\partial\theta}\log f(X,\theta)\cdot\left(\frac{\partial}{\partial\theta}\log f(X,\theta)\right)^T\right], \qquad X \sim P_\theta.$$

Rmk. This is the covariance matrix of

$$S_n(\theta) = \frac{\partial}{\partial\theta}\log f(X,\theta)$$

when $X \sim P_\theta$.

Prop 1. For all $\theta \in \text{int}(\Theta)$, in a "regular" model,

$$I(\theta) = -\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta\,\partial\theta^T}\log f(X,\theta)\right]$$

$$\left(I_{ij}(\theta) = -\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta_i\,\partial\theta_j}\log f(X,\theta)\right]\right).$$

Proof.

$$0 = \frac{\partial^2}{\partial\theta\,\partial\theta^T}\int f(x,\theta)\,dx = \int \frac{\partial^2}{\partial\theta\,\partial\theta^T} f(x,\theta)\,dx.$$

[When this equality holds, we say the model is "regular".]
Now, we have

$$\frac{\partial^2}{\partial\theta\,\partial\theta^T}\log f(x,\theta) = \frac{\partial}{\partial\theta^T}\left[\frac{1}{f(x,\theta)}\frac{\partial}{\partial\theta}f(x,\theta)\right]$$
$$= -\frac{1}{f(x,\theta)^2}\frac{\partial}{\partial\theta}f(x,\theta)\left(\frac{\partial}{\partial\theta}f(x,\theta)\right)^T + \frac{1}{f(x,\theta)}\frac{\partial^2}{\partial\theta\,\partial\theta^T}f(x,\theta).$$

Take $-\mathbb{E}_\theta[\cdot]$ of both sides.

$$-\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta\,\partial\theta^T}\log f(X,\theta)\right]$$
$$= \mathbb{E}_\theta\left[\frac{1}{f(X,\theta)^2}\frac{\partial}{\partial\theta}f(X,\theta)\left(\frac{\partial}{\partial\theta}f(X,\theta)\right)^T\right] + 0$$
$$= \mathbb{E}_\theta\left[\frac{\partial}{\partial\theta}\log f(X,\theta)\left(\frac{\partial}{\partial\theta}\log f(X,\theta)\right)^T\right] = I(\theta).$$

□

Tensorisation.

Suppose the observation X consists of n random variables $X_1, \ldots, X_n$. Then, the F.I. matrix is

$$I_n(\theta) = \mathbb{E}_\theta \left[\frac{\partial}{\partial \theta} \log f(X_1, \ldots, X_n, \theta) \left(\frac{\partial}{\partial \theta} \log f(X_1, \ldots, X_n, \theta) \right)^T \right].$$

If $X_1, \ldots, X_n$ are iid such that

$$f(X_1, \ldots, X_n, \theta) = \prod_{i=1}^{n} f(X_i, \theta),$$

where $\{f(x, \theta) : \theta \in \Theta\}$ is a model for X_1, then the F.I. is

$$\begin{aligned} I_n(\theta) &= \mathbb{E}_\theta \left[\sum_{i,j} \underbrace{\frac{\partial}{\partial \theta} \log f(X_i, \theta)}_{\substack{\text{mean 0 under} \\ P_\theta}} \left(\frac{\partial}{\partial \theta} \log f(X_j, \theta) \right)^T \right] \\ &= \mathbb{E}_\theta \left[\sum_{i=1}^{n} \frac{\partial}{\partial \theta} \log f(X_i, \theta) \left(\frac{\partial}{\partial \theta} \log f(X_i, \theta) \right)^T \right] \\ &= nI(\theta), \end{aligned}$$

where $I(\theta)$ is the Fisher Information for a single observation.

End of Lecture 2

Recap of Information Geometry

Observation 1.

$$\mathrm{KL}(P_{\theta_0}, P_\theta) = \mathbb{E}_{\theta_0} \left[\log \frac{f(X, \theta_0)}{f(X, \theta)} \right] \geq 0,$$

minimised at $\theta = \theta_0$.

Observation 2.

$$\mathbb{E}_\theta \left[\frac{\partial}{\partial \theta} \log f(X, \theta) \right] = 0, \qquad \forall \theta \in \mathrm{int}(\Theta).$$

Observation 3.

$$I(\theta) := \mathrm{Var}_\theta \left[\frac{\partial}{\partial \theta} \log f(X, \theta) \right] = -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta \, \partial \theta^T} \log f(X, \theta) \right], \qquad \forall \theta \in \mathrm{int}(\Theta).$$

Why geometry? The F.I., $I(\theta)$, defines a metric tensor on the manifold

$$\{f(\cdot, \theta) : \theta \in \Theta\}.$$

This is known as the **Fisher–Rao metric**. KL divergence is approximated locally by this metric.

- **Reparametrisation invariance.** Take the Bernoulli(θ) model, where θ is the probability of success. Reparametrise it in terms of "log-odds":

$$\eta = \log\left(\frac{\theta}{1-\theta}\right) \in \mathbb{R}.$$

  The manifold

$$\{f(\cdot, \eta) : \eta \in \mathbb{R}\},$$

  equipped with the metric tensor $I(\eta)$, preserves geodesics, path lengths, etc.

- **Sufficient statistic invariance.**

Chentsov's theorems. These properties characterise the Fisher–Rao metric.

Proposition 2 (Cramér–Rao Lower Bound)

Let

$$\{f(\cdot, \theta) : \theta \in \Theta \subseteq \mathbb{R}\}$$

be a "regular" statistical model. Let $\tilde{\theta}(X_1, \ldots, X_n)$ be an unbiased estimator based on i.i.d. samples $X_1, \ldots, X_n$:

$$\mathbb{E}_\theta \tilde{\theta} = \theta, \qquad \forall \theta \in \Theta.$$

Then

$$\mathrm{Var}_\theta(\tilde{\theta}) \geq \frac{1}{nI(\theta)}, \qquad \forall \theta \in \mathrm{int}(\Theta).$$

Proof. Recall the Cauchy–Schwarz inequality for random variables Y, Z:

$$\big(\mathbb{E}[YZ]\big)^2 \leq \mathbb{E}[Y^2]\mathbb{E}[Z^2].$$

Let

$$Z = \tilde{\theta} - \theta, \qquad Y = \frac{\partial}{\partial \theta} \log f(X_1, \ldots, X_n; \theta).$$

Then

$$\mathbb{E}[Z^2] = \mathrm{Var}_\theta(\tilde{\theta}), \qquad \text{as } \tilde{\theta} \text{ is unbiased,}$$

and

$$\mathbb{E}[Y^2] = nI(\theta), \qquad \text{as the F.I. tensorises.}$$

Moreover,

$$\begin{aligned}
\mathbb{E}[YZ] &= \mathbb{E}_\theta\left[\tilde{\theta}\frac{\partial}{\partial \theta} \log f(X_1, \ldots, X_n; \theta)\right] \\
&\quad - \theta \underbrace{\mathbb{E}_\theta\left[\frac{\partial}{\partial \theta} \log f(X_1, \ldots, X_n; \theta)\right]}_{=0} \\
&= \int_{\mathcal{X}^n} \tilde{\theta}(x_1, \ldots, x_n) \frac{\partial f(x_1, \ldots, x_n; \theta)}{\partial \theta} \frac{f(x_1, \ldots, x_n; \theta)}{f(x_1, \ldots, x_n; \theta)}\, dx_1 \cdots dx_n \\
&= \frac{\partial}{\partial \theta} \underbrace{\int_{\mathcal{X}^n} \tilde{\theta}\, f(x_1, \ldots, x_n; \theta)\, dx_1 \cdots dx_n}_{=\mathbb{E}_\theta \tilde{\theta} = \theta} \qquad \text{(by regularity)} \\
&= \frac{\partial \theta}{\partial \theta} = 1.
\end{aligned}$$

Plug these expectations into the Cauchy–Schwarz inequality and rearrange. □

Remark. Sometimes, the MLE is unbiased and attains the Cramér–Rao lower bound, but not always. However, we will show this is true asymptotically.

If $\Theta \subseteq \mathbb{R}^p$ with $p > 1$, let

$$\Phi : \Theta \to \mathbb{R}$$

be a scalar function of θ. Consider an unbiased estimator $\tilde{\Phi}(X_1, \dots, X_n)$. Then one can show that

$$\mathrm{Var}_\theta(\tilde{\Phi}) \geq \frac{1}{n} \left(\frac{\partial \Phi(\theta)}{\partial \theta} \right)^T I(\theta)^{-1} \frac{\partial \Phi(\theta)}{\partial \theta}, \qquad \forall \theta \in \mathrm{int}(\Theta).$$

Corollary 1. If $\tilde{\theta}$ is not unbiased, we have

$$\mathrm{Var}_\theta(\tilde{\theta}) \geq \frac{\dfrac{\partial}{\partial \theta} \mathbb{E}_\theta \tilde{\theta}}{nI(\theta)}, \qquad \forall \theta \in \mathrm{int}(\Theta).$$

Example. Take $\Phi(\theta) = \alpha^T \theta$ for some $\alpha \in \mathbb{R}^p$. In this case,

$$\frac{\partial \Phi(\theta)}{\partial \theta} = \alpha,$$

and

$$\mathrm{Var}_\theta(\tilde{\Phi}) \geq \frac{1}{n} \alpha^T I(\theta)^{-1} \alpha.$$

Example Sheet 1

We observe a single pair

$$(X_1, X_2) \sim N\left(\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix}, \Sigma \right),$$

where $\Sigma \succ 0$ is known. We wish to estimate θ_1.

Case 1. Assume that θ_2 is known. In this case, the model has one parameter and F.I. matrix

$$I_1(\theta_1).$$

Case 2. Assume that θ_2 is unknown. For any estimator

$$\tilde{\theta} = \begin{pmatrix} \tilde{\theta}_1 \\ \tilde{\theta}_2 \end{pmatrix},$$

we have

$$\tilde{\theta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}^T \tilde{\theta}.$$

So, if $\tilde{\theta}$ is unbiased, we have the Cramér–Rao lower bound

$$\mathrm{Var}_\theta(\tilde{\theta}_1) \geq \begin{pmatrix} 1 \\ 0 \end{pmatrix}^T I(\theta)^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \left[I(\theta)^{-1} \right]_{11}.$$

Goal. Show

$$I_1(\theta_1)^{-1} < \left[I(\theta)^{-1} \right]_{11},$$

unless Σ is diagonal (X_1, X_2 independent).

II. Asymptotic Theory of the MLE

All reasonable estimators are asymptotically unbiased:

$$\mathbb{E}_\theta \tilde{\theta} \xrightarrow{n\to\infty} \theta, \qquad \forall \theta \in \Theta.$$

We wish to prove that estimators are **consistent**:

$$\tilde{\theta} \xrightarrow{?} \theta \qquad \text{as } n \to \infty \quad \text{under } P_\theta, \qquad \forall \theta \in \Theta.$$

For consistent estimators, a reasonable notion of optimality is **asymptotic efficiency**:

$$n \operatorname{Var}_\theta(\tilde{\theta}) \xrightarrow{n\to\infty} I(\theta)^{-1}, \qquad \forall \theta \in \operatorname{int}(\Theta).$$

We will show that the MLE is asymptotically efficient and, furthermore,

$$\hat{\theta}_{\text{MLE}} \approx N\left(\theta, n^{-1} I(\theta)^{-1}\right), \qquad \forall \theta.$$

This distributional convergence can be used for statistical inference.

Stochastic Convergence

Definition 1. Let $(X_n : n \in \mathbb{N})$ be a collection of random variables in some probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- We say that X_n converges almost surely to a random variable X on $(\Omega, \mathcal{F}, \mathbb{P})$, or

  $$X_n \xrightarrow{\text{a.s.}} X,$$

  if and only if

  $$\mathbb{P}\left(\left\{\omega : \|X_n(\omega) - X(\omega)\|_2 \xrightarrow{n\to\infty} 0\right\}\right) = 1.$$

- We say that X_n converges to X in probability, or

  $$X_n \xrightarrow{P} X,$$

  if, for every $\varepsilon > 0$,

  $$\mathbb{P}\left(\|X_n - X\|_2 > \varepsilon\right) \xrightarrow{n\to\infty} 0.$$

End of Lecture 3

Recall: If $(X_n), X$ are random vectors in $\mathbb{R}^k$, we say

- $X_n \xrightarrow{\text{a.s.}} X$ iff

  $$\mathbb{P}\left(\|X_n - X\|_2 \xrightarrow{n\to\infty} 0\right) = 1.$$

- $X_n \xrightarrow{P} X$ iff

  $$\forall \varepsilon > 0, \qquad \mathbb{P}\left(\|X_n - X\|_2 > \varepsilon\right) \xrightarrow{n\to\infty} 0.$$

Rmk.

$$X_n \xrightarrow{\text{a.s./}P} X \quad \text{iff} \quad X_{n,i} \xrightarrow{\text{a.s./}P} X_i \qquad \forall\, 1 \le i \le k.$$

$$[X \le t \iff X_i \le t_i \quad \forall\, 1 \le i \le k]$$

Def 2. We say that X_n converges in distribution to X, or $X_n \xrightarrow{d} X$, if for all $t \in \mathbb{R}^k$ where the function

$$t \mapsto \mathbb{P}(X \leq t)$$

is continuous, we have

$$\mathbb{P}(X_n \leq t) \xrightarrow{n\to\infty} \mathbb{P}(X \leq t).$$

Some facts from P & M.

Prop 1.

(a)

$$X_n \xrightarrow{\text{a.s.}} X \implies X_n \xrightarrow{P} X \implies X_n \xrightarrow{d} X.$$

(b) Continuous Mapping Theorem (CMT). Let $(X_n), X$ take values in $\mathcal{X} \subseteq \mathbb{R}^k$, let $g : \mathcal{X} \to \mathbb{R}$ be continuous. Then,

$$X_n \xrightarrow{\text{a.s.}/P/d} X \implies g(X_n) \xrightarrow{\text{a.s.}/P/d} g(X).$$

(c) Slutsky's Lemma. Suppose that

$$X_n \xrightarrow{d} X, \qquad Y_n \xrightarrow{d} c,$$

where c is constant, then

(i)

$$Y_n \xrightarrow{P} c.$$

(ii)

$$X_n + Y_n \xrightarrow{d} X + c.$$

(iii) If $k = 1$,

$$X_n Y_n \xrightarrow{d} X \cdot c,$$
$$\frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{c} \qquad \text{if } c \neq 0.$$

(iv) Let (A_n) be a sequence of random matrices in $\mathbb{R}^{m\times k}$, and suppose,

$$(A_n)_{ij} \xrightarrow{P} A_{ij} \qquad \forall i, j$$

where A is deterministic. Then,

$$A_n X_n \xrightarrow{d} AX.$$

(d) If $X_n \xrightarrow{d} X$, then (X_n) is "bounded in probability" or $X_n = O_P(1)$, i.e. for all $\varepsilon > 0$, $\exists\, M_\varepsilon > 0$, $N_\varepsilon \in \mathbb{N}$ s.t.

$$\mathbb{P}\left(\|X_n\|_2 > M_\varepsilon\right) < \varepsilon \qquad \forall n \geq N_\varepsilon.$$

2. LLN & CLT

Let $X_1, X_2, \ldots$ be iid copies of a r.v. X in $\mathbb{R}^k$. When $k = 1$, the Weak Law of large numbers says that if $\text{Var}(X) < \infty$, then

$$\bar{X}_n := \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{P} \mathbb{E}X.$$

Indeed by Chebyshev's inequality,

$$\forall \varepsilon > 0, \qquad \mathbb{P}\left(|\bar{X}_n - \mathbb{E}X| > \varepsilon\right) \leq \frac{\mathrm{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\mathrm{Var}(X)}{n\varepsilon^2} \xrightarrow{n\to\infty} 0.$$

Theorem 1 (Strong law of large numbers). If $\mathbb{E}\left(\|X_n\|_2\right) < \infty$, then

$$\bar{X}_n \xrightarrow{\text{a.s.}} \mathbb{E}[X] \qquad \text{as } n \to \infty.$$

Theorem 2 (Central Limit Theorem). When $k = 1$, assume in addition that $\mathrm{Var}(X) = \sigma^2 < \infty$, then,

$$\sqrt{n}\left(\bar{X}_n - \mathbb{E}X\right) \xrightarrow{d} N(0, \sigma^2).$$

To state a multivariate version, recall that $Z \sim N(\mu, \Sigma)$ iff

$$\forall \alpha \in \mathbb{R}^k, \qquad \alpha^T Z \sim N(\alpha^T \mu, \alpha^T \Sigma \alpha).$$

If Σ is positive definite, then Z has a p.d.f.

$$f(z) = \frac{1}{(2\pi)^{n/2}} \frac{1}{|\det \Sigma|^{1/2}} \exp\left\{-\frac{1}{2}(z-\mu)^T \Sigma^{-1}(z-\mu)\right\}, \qquad z \in \mathbb{R}^k.$$

Theorem 2 ($k \geq 1$). If $\mathrm{Var}(X) = \Sigma$ is finite & pos. def. then

$$\sqrt{n}\left(\bar{X}_n - \mathbb{E}X\right) \xrightarrow{d} N(0, \Sigma).$$

Cor. By Prop 1(d),

$$\sqrt{n}\left(\bar{X}_n - \mathbb{E}X\right) = O_P(1) \iff \left(\bar{X}_n - \mathbb{E}X\right) = O_P\left(\frac{1}{\sqrt{n}}\right).$$

The CLT is useful to define confidence intervals. Let $X_1, X_2, \ldots$ be iid copies of X with mean μ_0 and $\mathrm{Var}(X) = \sigma^2 < \infty$ known. Define

$$C_n = \left\{\mu \in \mathbb{R} : \sqrt{n}\frac{|\bar{X}_n - \mu|}{\sigma} \leq z_\alpha\right\}$$

where z_α is such that

$$\mathbb{P}(|Z| \leq z_\alpha) = 1 - \alpha \qquad \text{if } Z \sim N(0,1).$$

$$\begin{aligned}
\mathbb{P}(\mu_0 \in C_n) &= \mathbb{P}\left(\sqrt{n}|\bar{X}_n - \mu_0| \leq \sigma z_\alpha\right) \\
&\xrightarrow{n\to\infty} \mathbb{P}(|Z| \leq z_\alpha) \qquad \text{by the CLT} \\
&= 1 - \alpha.
\end{aligned}$$

Hence, we say that C_n is an "asymptotic" $(1-\alpha)$-confidence interval.

If σ^2 is unknown, we can define an estimator

$$\begin{aligned}
\hat{\sigma}^2 &= \frac{1}{n-1}\sum_{i=1}^n \left(X_i - \underbrace{\bar{X}_n}_{\frac{1}{n}\sum_j X_j}\right)^2 \\
&= \frac{1}{n-1}\left(\sum_{i=1}^n X_i^2 - n\bar{X}_n^2\right) \\
&\xrightarrow{d} \mathbb{E}(X^2) - (\mathbb{E}X)^2 = \sigma^2
\end{aligned}$$

by several applications of the SLLN, the CMT and Slutsky's lemma.

Define a new random interval

$$\widetilde{C}_n = \left\{ \mu \in \mathbb{R} : \sqrt{n} \frac{|\bar{X}_n - \mu|}{\hat{\sigma}} \leq z_\alpha \right\}.$$

By the CMT, $\hat{\sigma} \xrightarrow{d} \sigma$ constant. By Slutsky's Lemma,

$$\sqrt{n} \frac{(\bar{X}_n - \mu_0)}{\hat{\sigma}} \xrightarrow{d} N(0,1).$$

Hence, C_n is an asymptotic $(1-\alpha)$-confidence interval for μ_0.

Next time: Uniform laws of Large Numbers (ULLN)

Let $\mathcal{H}$ be a class of functions $\mathbb{R}^k \to \mathbb{R}$. Assume that

$$\mathbb{E}|h(X)| < \infty \qquad \forall h \in \mathcal{H}.$$

We know from the SLLN that

$$\frac{1}{n} \sum_{i=1}^{n} h(X_i) \xrightarrow{\text{a.s.}} \mathbb{E}[h(X)] \qquad \forall h \in \mathcal{H}.$$

We have a ULLN if

$$\sup_{h \in \mathcal{H}} \left| \frac{1}{n} \sum_{i=1}^{n} h(X_i) - \mathbb{E}[h(X)] \right| \xrightarrow{\text{a.s.}} 0.$$

End of Lecture 4

Uniform Laws of Large Numbers

Let X be a r.v. on $\mathcal{X}$. (X_i) iid copies of X. Let $\mathcal{H}$ be a class of functions $\mathcal{X} \to \mathbb{R}$, s.t.

$$\mathbb{E}|h(X)| < \infty \qquad \forall h \in \mathcal{H}.$$

$$\text{ULLN} \qquad \sup_{h \in \mathcal{H}} \left| \underbrace{\frac{1}{n} \sum_{i=1}^{n} h(X_i)}_{=P_n h} - \underbrace{\mathbb{E}h(X)}_{=Ph} \right| \xrightarrow{\text{a.s.}} 0.$$

If $\mathcal{H} = \{h_1, \ldots, h_M\}$ is finite, then by the SLLN, for any $1 \leq j \leq M$,

$$|P_n h_j - P h_j| \xrightarrow{\text{a.s.}} 0$$

i.e. on an event A_j with $\mathbb{P}(A_j) = 1$, $\forall \varepsilon > 0$, $\exists N_j(\varepsilon)$ s.t.

$$|P_n h_j - P h_j| < \varepsilon \qquad \forall n > N_j(\varepsilon).$$

Hence on $A := \bigcap_{j=1}^{M} A_j$, we have $\forall \varepsilon > 0$,

$$\max_{1 \leq j \leq M} |P_n h_j - P h_j| < \varepsilon \qquad \forall n > \max_j N_j(\varepsilon).$$

As $\mathbb{P}(A) = 1$, this means

$$\sup_{h \in \mathcal{H}} |P_n h - P h| \xrightarrow{\text{a.s.}} 0.$$

When $|\mathcal{H}|$ is infinite, we will use a "bracketing argument". A bracket is a collection of functions $\{\underline{h}_j, \overline{h}_j\}_{j=1}^{M}$ such that

$$\forall j, \qquad \underline{h}_j(x) \leq \overline{h}_j(x) \qquad \forall x \in \mathcal{X}.$$

A finite bracket covers $\mathcal{H}$ if for all $h \in \mathcal{H}$, $\exists j \in \{1, \ldots, M\}$ s.t.

$$\underline{h}_j(x) \leq h(x) \leq \overline{h}_j(x) \qquad \forall x \in \mathcal{X}.$$

Prop 2. Suppose that $\forall \varepsilon > 0$, there is a finite bracket $\{\underline{h}_j, \overline{h}_j\}_{j=1}^M$ covering $\mathcal{H}$, s.t.

$$(1) \quad \mathbb{E}|\underline{h}_j(X)| < \infty, \quad \mathbb{E}|\overline{h}_j(X)| < \infty \qquad \forall 1 \le j \le M,$$
$$(2) \quad \mathbb{E}|\overline{h}_j(X) - \underline{h}_j(X)| < \varepsilon.$$

Then,

$$\sup_{h \in \mathcal{H}} |P_n h - Ph| \xrightarrow{\text{a.s.}} 0.$$

Proof. Take any $\varepsilon > 0$, then for any $h \in \mathcal{H}$, $\exists 1 \le j \le M$ s.t. $\underline{h}_j \le h \le \overline{h}_j$. Then

$$\begin{aligned} P_n h - Ph &\le P_n \overline{h}_j - P\underline{h}_j \\ &\le |P_n \overline{h}_j - P\overline{h}_j| + \underbrace{|P(\overline{h}_j - \underline{h}_j)|}_{\le \varepsilon}. \end{aligned}$$

Similarly,

$$Ph - P_n h \le P\overline{h}_j - P_n \underline{h}_j \le |P_n \underline{h}_j - P\underline{h}_j| + \underbrace{|P(\overline{h}_j - \underline{h}_j)|}_{\le \varepsilon}.$$

$$\therefore \quad \sup_{h \in \mathcal{H}} |P_n h - Ph| \le \underbrace{\max_{g \in \text{bracket}} |P_n g - Pg|}_{\xrightarrow{\text{a.s.}} 0 \text{ by the SLLN as bracket is finite}} + \varepsilon.$$

Hence, on an event A_ε with $\mathbb{P}(A_\varepsilon) = 1$, we have

$$\limsup_{n \to \infty} \sup_{h \in \mathcal{H}} |P_n h - Ph| \le \varepsilon.$$

Take sequence $\varepsilon_k \downarrow 0$. Let $A = \bigcap_{k=1}^\infty A_{\varepsilon_k}$, with $\mathbb{P}(A) = 1$. On A,

$$\sup_{h \in \mathcal{H}} |P_n h - Ph| \longrightarrow 0. \qquad \square$$

3 Consistency of the MLE

Def 3. We say an estimator $\hat{\theta}(X_1, \ldots, X_n)$ is consistent if $\hat{\theta} \xrightarrow{P} \theta$ under P_θ $\forall \theta \in \Theta$.

Assumptions. Let $\{f(\cdot, \theta) : \theta \in \Theta\}$ be a model of pdfs or pmfs on $\mathcal{X} \subseteq \mathbb{R}^d$. We'll assume

(1) $f(x, \theta) > 0 \qquad \forall x \in \mathcal{X},\ \theta \in \Theta$.

(2) $\theta \mapsto f(x, \theta)$ is continuous $\forall x \in \mathcal{X}$.

(3) Θ is compact.

(4) Strict identifiability: $f(\cdot, \theta) = f(\cdot, \theta') \implies \theta = \theta'$.

(5) $\mathbb{E}_\theta \underbrace{\sup_{\theta' \in \Theta} |\log f(X, \theta')|}_{M(X)} < \infty \qquad \forall \theta \in \Theta$.

Recall that (5) allows us to define $\ell(\theta) := \mathbb{E}_{\theta_0} \log f(X, \theta)$ for a true parameter θ_0. Furthermore, by (4) $\ell(\theta)$ is maximised uniquely at $\theta = \theta_0$.

By (2) & (5), we have that ℓ is continuous. Indeed take (θ_i), $\theta_i \to \theta$, then

$$|\ell(\theta_i) - \ell(\theta)| = \left| \mathbb{E}_{\theta_0} \left[\underbrace{\log f(X, \theta_i) - \log f(X, \theta)}_{\xrightarrow{\text{a.s.}} 0 \text{ by (2)}} \right] \right|$$
$$\longrightarrow 0 \quad \text{by the Dominated Convergence Theorem (P\&M) using the domination condition (5).}$$

In maximum likelihood we maximise

$$\bar{\ell}_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} \log f(X_i, \theta), \qquad \text{a "sample" version of } \ell(\theta).$$

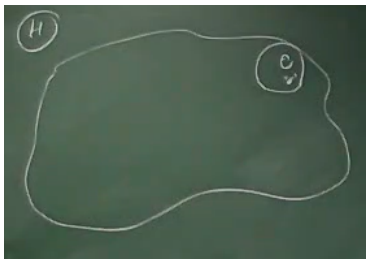
Lemma. If (X_i) are iid with law P_{θ_0}, we have the ULLN

$$\sup_{\theta\in\Theta} \left| \underbrace{\bar{\ell}_n(\theta)}_{=P_n h_\theta} - \underbrace{\ell(\theta)}_{=Ph_\theta} \right| \xrightarrow{\text{a.s.}} 0$$

where $h_\theta : x \mapsto \log f(x, \theta)$. Here $\mathcal{H} = \{h_\theta : \theta \in \Theta\}$.

Proof sketch To apply Prop 2, we need to construct for any $\varepsilon > 0$ a finite bracket satisfying the hypotheses (1) & (2). Take any $\theta \in \Theta$, let $B_{\theta,\delta} = \{\theta' : \|\theta - \theta'\| \le \delta\}$, let

$$\overline{h}_\theta^{(\delta)} = \sup_{\theta'\in B_{\theta,\delta}} h_{\theta'}(x), \qquad \underline{h}_\theta(x) = \inf_{\theta'\in B_{\delta,\theta}} h_{\theta'}(x).$$



By def

$$\underline{h}_\theta \le h_{\theta'} \le \overline{h}_\theta \qquad \forall \theta' \in B_{\delta,\theta}.$$

Hypothesis (1) is given by Assumption (5). We'll choose δ small enough so that we can obtain Hypothesis 2:

$$\mathbb{E}|\overline{h}_\theta(X) - \underline{h}_\theta(X)| < \varepsilon$$

(To do this carefully, apply the DCT using assumptions (2) & (5).)

For each $\theta \in \Theta$ we obtain a pair $(\underline{h}_\theta, \overline{h}_\theta)$ sandwiching functions in a "ball" around h_θ. By compactness, $\exists$ a finite bracket $\{\overline{h}_{\theta_j}, \underline{h}_{\theta_j}\}_{j=1}^{M}$ s.t. $\forall \theta \in \Theta$, $\exists j$ s.t.

$$\underline{h}_{\theta_j} \le h_\theta \le \overline{h}_{\theta_j}.$$

This allows us to apply prop 2. $\square$

End of Lecture 5

Recall. We stated 5 assumptions for consistency of the MLE.
Implications

- $\ell(\theta) = \mathbb{E}_{\theta_0}[\log f(X, \theta)] < \infty \ \forall \theta$, maximised uniquely at θ_0.
- ℓ is continuous on Θ.
- ULLN

$$\sup_{\theta\in\Theta} |\bar{\ell}_n(\theta) - \ell(\theta)| \xrightarrow{\text{a.s./}P/d} 0.$$

Theorem 3. Suppose that $\{f(\cdot,\theta) : \theta \in \Theta\}$ satisfies the stated assumptions and $X_1, X_2, \ldots$ are iid P_{θ_0}. Then an MLE $\hat{\theta}_n$ based on $X_1, \ldots, X_n$ exists and

$$\hat{\theta}_n \xrightarrow{P} \theta_0.$$

Proof.

$$\bar{\ell}_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} \log f(X_i, \theta)$$

is continuous, so it attains a supremum in Θ compact at some $\hat{\theta}_n \in \Theta$.

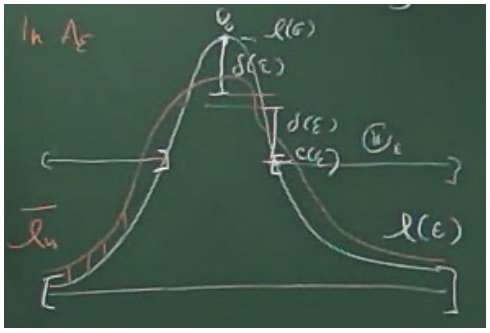
Define

$$\Theta_\varepsilon = \{\theta \in \Theta : \|\theta - \theta_0\|_2 \geq \varepsilon\}$$

and note Θ_ε is compact (we remove an open ball from Θ compact). We'll show that

$$\mathbb{P}(\hat{\theta}_n \in \Theta_\varepsilon) \longrightarrow 0 \qquad \text{as } n \to \infty.$$

If this holds for arbitrary $\varepsilon > 0$, $\hat{\theta}_n \xrightarrow{P} \theta_0$.



$$\sup_{\theta \in \Theta_\varepsilon} \ell(\theta) =: c(\varepsilon) < \ell(\theta_0).$$

Choose

$$\delta(\varepsilon) < \frac{\ell(\theta_0) - c(\varepsilon)}{2}.$$

Define an event

$$A_\varepsilon = \left\{ \sup_{\theta \in \Theta} |\bar{\ell}_n(\theta) - \ell(\theta)| \leq \delta(\varepsilon) \right\}.$$

By the ULLN, $\mathbb{P}(A_\varepsilon) \to 1$ as $n \to \infty$.

$$\begin{aligned}
\sup_{\theta \in \Theta_\varepsilon} \bar{\ell}_n(\theta) &\leq \sup_{\theta \in \Theta_\varepsilon} \ell(\theta) + \underbrace{\sup_{\theta \in \Theta_\varepsilon} (\bar{\ell}_n(\theta) - \ell(\theta))}_{\leq \delta(\varepsilon) \text{ on } A_\varepsilon} \\
&\leq c(\varepsilon) + \delta(\varepsilon) \\
&< \ell(\theta_0) - \delta(\varepsilon) \qquad \text{(by def of } \delta(\varepsilon)) \\
&\leq \bar{\ell}_n(\theta_0).
\end{aligned}$$

Hence, on A_ε, $\hat{\theta}_n \notin \Theta_\varepsilon$. In other words $A_\varepsilon \subseteq \{\hat{\theta}_n \notin \Theta_\varepsilon\}$. Therefore

$$\mathbb{P}(\hat{\theta}_n \notin \Theta_\varepsilon) \geq \mathbb{P}(A_\varepsilon) \longrightarrow 1 \qquad \text{as } n \to \infty.$$

Finally

$$\mathbb{P}(\hat{\theta}_n \in \Theta_\varepsilon) \longrightarrow 0 \qquad \text{as } n \to \infty. \qquad \square$$

Rmk. If we assume that $\theta \mapsto \log f(x,\theta)$ is differentiable for all $x \in \mathcal{X}$, and $\frac{\partial \bar{\ell}_n(\theta)}{\partial \theta}$ satisfies a ULLN and its limit $\frac{\partial \ell(\theta)}{\partial \theta}$ has a single 0 at θ_0, there is a simpler argument to show consistency. Can deal with noncompact Θ in some simple models.

4. Asymptotic Normality of the MLE

Fluctuations of $\sqrt{n}(\hat{\theta}_n - \theta_0)$.

Theorem 4. Suppose that $\{f(\cdot,\theta) : \theta \in \Theta\}$ satisfies the 9 stated assumptions. Let $\hat{\theta}_n$ be an MLE based on $X_1, \ldots, X_n$ iid P_{θ_0}. Then as $n \to \infty$,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, I^{-1}(\theta_0)).$$

The nine assumptions are the five assumptions for consistency stated above, together with the following four additional regularity assumptions (R2).

Regularity assumptions (R2). Let the statistical model be such that:

(1) $\theta_0 \in \operatorname{int} \Theta$;

(2) there exists an open neighbourhood N of θ_0 on which $\theta \mapsto f(x,\theta)$ is twice continuously differentiable for all $x \in \mathcal{X}$;

(3) $I(\theta_0)$ exists and is invertible;

(4)
$$\int_{\mathcal{X}} \sup_{\theta \in \Theta} \|\nabla_\theta \log f(x,\theta)\|_2 \, dx < \infty$$
$$\mathbb{E}_{\theta_0}\left(\sup_{\theta \in \Theta} \|\nabla^2_\theta \log f(X,\theta)\|_2 \right) < \infty$$
$$\int_{\mathcal{X}} \sup_{\theta \in \Theta} \|\nabla^2_\theta \log f(x,\theta)\|_2 \, dx < \infty.$$

(In fact the above can be weakened by replacing Θ above with any compact set K with non-empty interior that contains θ_0.)

Proof. Define $\varepsilon > 0$ s.t. the ball

$$\{\theta \in \Theta : \|\theta - \theta_0\| < \varepsilon\} \subseteq K \subset U.$$

Here K is a compact set with non-empty interior containing θ_0, as in (R2) above; U denotes the open neighbourhood called N in (R2).

We'll work in an event

$$E_n = \{\|\hat{\theta}_n - \theta_0\| < \varepsilon\}.$$

By consistency, $\mathbb{P}(E_n) \to 1$ as $n \to \infty$. We'll show that

$$\mathbf{1}_{E_n}\sqrt{n}(\hat{\theta}_n - \theta_0) = \mathbf{1}_{E_n} W_n \xrightarrow{d} N(0, I^{-1}(\theta_0)) \qquad \text{(Ex sheet 1, 8(a))}.$$

Throughout the proof, just assume we're in E_n.

On E_n, the regularity conditions guarantee that

$$\nabla_\theta \bar{\ell}_n(\hat{\theta}_n) = 0.$$

The MVT implies that

$$0 = \nabla_\theta \overline{\ell}_n(\hat{\theta}_n) = \nabla_\theta \overline{\ell}_n(\theta_0) + \overline{A}_n(\hat{\theta}_n - \theta_0),$$

where $\overline{A}_n \approx \nabla^2_\theta \overline{\ell}_n(\theta_0)$ is given by

$$(\overline{A}_n)_{i,j} = \frac{\partial^2}{\partial\theta_i \partial\theta_j} \overline{\ell}_n(\theta^{(i)}) \qquad \text{for } \theta^{(i)} \in [\theta_0, \hat{\theta}_n].$$

Rearranging yields

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = (-\overline{A}_n)^{-1} \sqrt{n}\, \nabla_\theta \overline{\ell}_n(\theta_0)$$

(reg conditions imply $\overline{A}_n$ is invertible).

Claim.

$$\overline{A}_n \xrightarrow{P} \mathbb{E}_{\theta_0}[\nabla^2_\theta \overline{\ell}_n(\theta_0)] = -I(\theta_0).$$

Cor. By the CMT,

$$\overline{A}_n^{-1} \xrightarrow{P} -I(\theta_0)^{-1}.$$

Furthermore,

$$\sqrt{n}\, \nabla_\theta \overline{\ell}_n(\theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \left(\underbrace{\nabla_\theta \log f(X_i, \theta_0)}_{\text{iid}} - \underbrace{\mathbb{E}_{\theta_0}[\nabla_\theta \log f(X, \theta_0)]}_{=0 \text{ (reg)}} \right).$$

By the multivariate CLT,

$$\sqrt{n}\, \nabla_\theta \overline{\ell}_n(\theta_0) \xrightarrow{d} N\left(0, \underbrace{\mathrm{Var}_{\theta_0}(\nabla_\theta \log f(X, \theta_0))}_{I(\theta_0)}\right).$$

By Slutsky's Lemma we have

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N\left(0, I(\theta_0)^{-1} I(\theta_0) I(\theta_0)^{-1}\right). \qquad \square$$

Next time: Proving claim

End of lecture 6